Automating the Fragmentation of Proteins

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Introduction

The fragmentation of proteins provides a way to approximate the energy of large systems through the summation of fragment energies within that system using the many-body expansion (MBE). As calculations need to be run on systems with thousands of atoms, MBE is necessary to reduce time complexity. However, the approximation is dependent on the fragments being used. Ideally, fragments will be large enough to maintain accuracy, but small enough so that they can be run efficiently. This project aims to test fragmentation by amino acid within proteins, as well as provide the groundwork for the training of ML approaches to fragmentation.

Many-Body Expansion

Many-body expansion is a method to perform expensive energy calculations by reducing a molecular system into fragments that can be calculated in more efficiently. However, MBE typically needs corrections to accelerate convergence [1].

$$E_{\text{tot}} \approx E^{(n)}_{\text{cb-MBE}} = \sum_I E^{(1)}_I + \sum_{I<J} \Delta E^{(2)}_{IJ} + \sum_{I<J<K} \Delta E^{(3)}_{IJK} + \cdots + \sum \Delta E^{(n)}_{IJK\ldots}$$

Figure 1. Many Body Expansion without corrections [2]

Why PDB files?

The Protein Data Bank hosts over 239,000 protein structures. Coordinates are stored in legacy files that must be parsed. However, these files contain where amino acids are within the protein and the atoms that belong to them (Figure 2). This makes fragmentation by amino acid trivial.

```
ATOM     81  N   PRO A   5       2.356   0.585 -10.838  1.00  0.72           N
ANISOU   81  N   PRO A   5      110     81     65    -16     59      6       N
ATOM     82  CA  PRO A   5       3.830   0.609 -10.874  1.00  1.06           C
ANISOU   82  CA  PRO A   5       94    137    149     61     59     82       C
ATOM     83  C   PRO A   5       4.425   1.987 -11.057  1.00  1.30           C
ANISOU   83  C   PRO A   5       84    171    211     34     23     60       C
ATOM     84  O   PRO A   5       5.622   2.139 -10.839  1.00  3.97           O
ANISOU   84  O   PRO A   5       79    396    944    -13    -74    213       O
ATOM     85  CB  PRO A   5       4.152  -0.297 -12.070  1.00  2.50           C
ANISOU   85  CB  PRO A   5      278    201    416    109    235    -21       C
ATOM     86  CG  PRO A   5       2.960  -0.138 -12.986  1.00  2.59           C
ANISOU   86  CG  PRO A   5      408    363    154    -37    203   -110       C
ATOM     87  CD  PRO A   5       1.788  -0.108 -12.012  1.00  1.37           C
ANISOU   87  CD  PRO A   5      254    160     76    -32     81    -56       C
ATOM     88  HA  PRO A   5       4.183   0.207 -10.053  1.00  1.28           H
ATOM     89  HB2 PRO A   5       4.968  -0.012 -12.510  1.00  3.00           H
ATOM     90  HB3 PRO A   5       4.252  -1.220 -11.788  1.00  3.00           H
ATOM     91  HG2 PRO A   5       3.015   0.686 -13.494  1.00  3.11           H
ATOM     92  HG3 PRO A   5       2.887  -0.886 -13.599  1.00  3.11           H
ATOM     93  HD2 PRO A   5       1.038   0.384 -12.380  1.00  1.64           H
ATOM     94  HD3 PRO A   5       1.498  -1.005 -11.783  1.00  1.64           H
ATOM     95  N   SER A   6       3.634   2.973 -11.482  1.00  1.01           N
ANISOU   95  N   SER A   6       55     86    218    -19      2     62       N
ATOM     96  CA  SER A   6       4.103   4.328 -11.657  1.00  1.00           C
ANISOU   96  CA  SER A   6       71     78    207    -29     23     -4       C
ATOM     97  C   SER A   6       2.892   5.258 -11.563  1.00  0.93           C
ANISOU   97  C   SER A   6      107     89    136     -6     24    -19       C
ATOM     98  O   SER A   6       1.743   4.820 -11.639  1.00  1.59           O
ANISOU   98  O   SER A   6       85    171    313    -31     40    -21       O
ATOM     99  CB  SER A   6       4.823   4.533 -12.984  1.00  1.82           C
ANISOU   99  CB  SER A   6      165    145    339      2    159     31       C
ATOM    100  OG  SER A   6       3.880   4.487 -14.023  1.00  3.06           O
ANISOU  100  OG  SER A   6      375    495    224    -14    110     14       O
ATOM    101  H   SER A   6       2.812   2.796 -11.659  1.00  1.21           H
ATOM    102  HA  SER A   6       4.722   4.543 -10.928  1.00  1.20           H
ATOM    103  HB2 SER A   6       5.274   5.391 -12.986  1.00  2.18           H
ATOM    104  HB3 SER A   6       5.488   3.838 -13.111  1.00  2.18           H
ATOM    105  HG  SER A   6       3.316   3.915 -13.855  1.00  4.60           H
```

Figure 2. Example PDB Layout

Criteria for Adding Caps

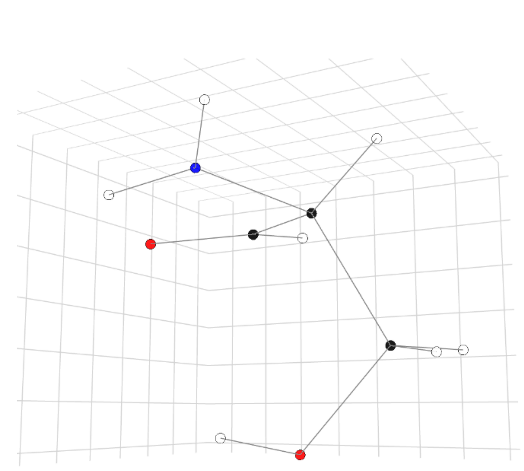

Figure 3. 3D Model of Serine

The existing method for establishing bonds relies on how distances relate to atomic radii. However, this method does not differentiate between single, double, or triple bonds. PDB files don't always contain correct connectivity, and should only be used for criteria as a last resort. Instead, bond angles and electronegativity should have more influence.

Methods and Code Development

Developing a file parser is trivial, but running energy calculations based on the amino acid data present will yield unreasonable results. Because the approximation of proteins is based on many-body expansion, some further correction must be made [1]. For this project, these corrections involved the attachment of hydrogen to atoms whose bonds were severed in fragmentation. In addition to operating as a file parser, any code written should also ensure the fragments' charge is balanced. To do so, some key functions were developed:

- Establish AA bond matrix Within a molecule, establish bonds between atoms using adjacency matrices and distance thresholds
- Identify bond types The bond matrix only contains information about if a bond exists, not its type.
- Locate atoms with missing bonds Given the bonds established in the above step, record indices of atoms with a positive or negative charge.
- Identify bonds cut From the list of atoms with less bonds than needed, find out which formed bonds with other amino acids. Then, take the coordinates of the atom that was connected, and attach a hydrogen along a vector.

Application

The bulk of the complexity in this code comes in creating bond and adjacency matrices ($O(n^2)$, where n represents the number of atoms in a protein). Capping and reading the PDB file adds negligible complexity, making the program efficient in writing .xyz files. Previously, this process had to be done by hand, While not terrible for small peptides, it became unruly considering the size of proteins. The capping for fragments also proves successful, only adding caps if there were bonds cut. This means that structures within the protein are not altered, even if they had a negative or positive charge.

In creating fragments, the code also outputs a "Cuts_made" text file which contains the indices of atoms within the protein whose bonds were cut to create a fragment. If fragmentation by amino acid proves to be successful, this means that each run can be used as training data.

The way functions are structured within the code means they can easily be exported for use in other problems. For example, the functions in "bond_info.py" can be used to check for extra charges within molecules before submitting to energy calculations [3].

Future Development and ML Pipeline

The code discussed in this poster was developed as a way to fill in some gaps for the Machine Learning Pipeline in line SIMCODES' goals [4]. More specifically, ML models may catch patterns in generating fragments that we aren't able to see.

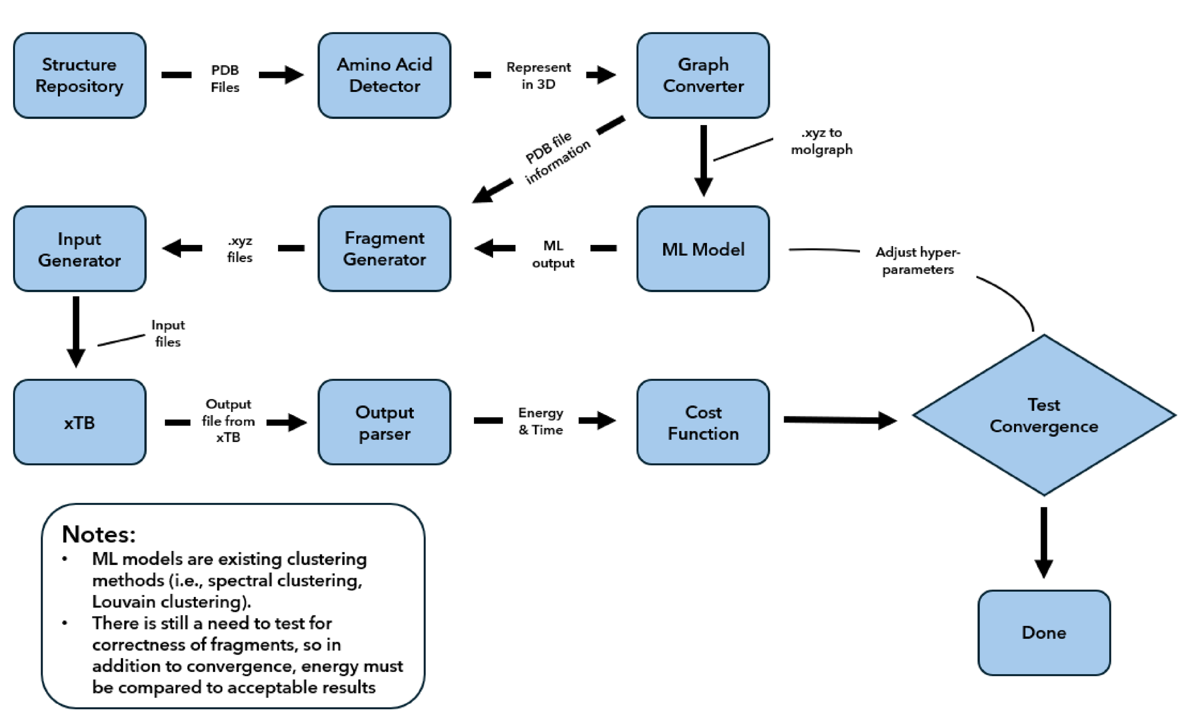

Figure 4. Machine Learning Pipeline

This code should be used to study fragmentation by amino acid. Many-body expansion with small fragments sacrifices accuracy in favor of reduced complexity. This code can be adapted to change which amino acids are included in a fragment, thus providing a way to find ideal fragment sizes. This would be beneficial in creating training data. Further, empirical study would be useful in understanding the exact relation between size and accuracy.

Additionally, the code needs improvement in representing metallic properties. Since there are limited kinds of atoms in amino acids, transition metals were not considered in this project. However, one long term goal of SIMCODES is to simulate metal-protein interactions. In order to align with this, the criteria for bond formation should be expanded to consider metals and their oxidation states.

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References

[1] Kuan-Yu Liu and John M. Herbert. "Understanding the many-body expansion for large systems. III. Critical role of four-body terms, counterpoise corrections, and cutoffs". In: *The Journal of Chemical Physics* 147.16 (Sept. 2017), p. 161729. ISSN: 0021-9606. DOI: 10.1063/1.4986110. eprint: https://pubs.aip.org/aip/jcp/article-pdf/doi/10.1063/1.4986110/15537025/161729_1_online.pdf. URL: https://doi.org/10.1063/1.4986110.

[2] Stefanie Schürmann et al. "Accurate quantum-chemical fragmentation calculations for ion–water clusters with the density-based many-body expansion". In: *Phys. Chem. Chem. Phys.* 25 (1 2023), pp. 736–748. DOI: 10.1039/D2CP04539G. URL: http://dx.doi.org/10.1039/D2CP04539G.

[3] Gabriela Campbell. *PDB_FileMaker*. Version 1.0.0. July 2025. URL: https://github.com/SIMCODES-ISU/Campbell_Repo/tree/main/PDB_FileMaker.

[4] Ryan Richard. "Automating the Fragmentation of Proteins". In: (July 2025). URL: https://simcodes-isu.github.io/projects/fragmenting_protein/.